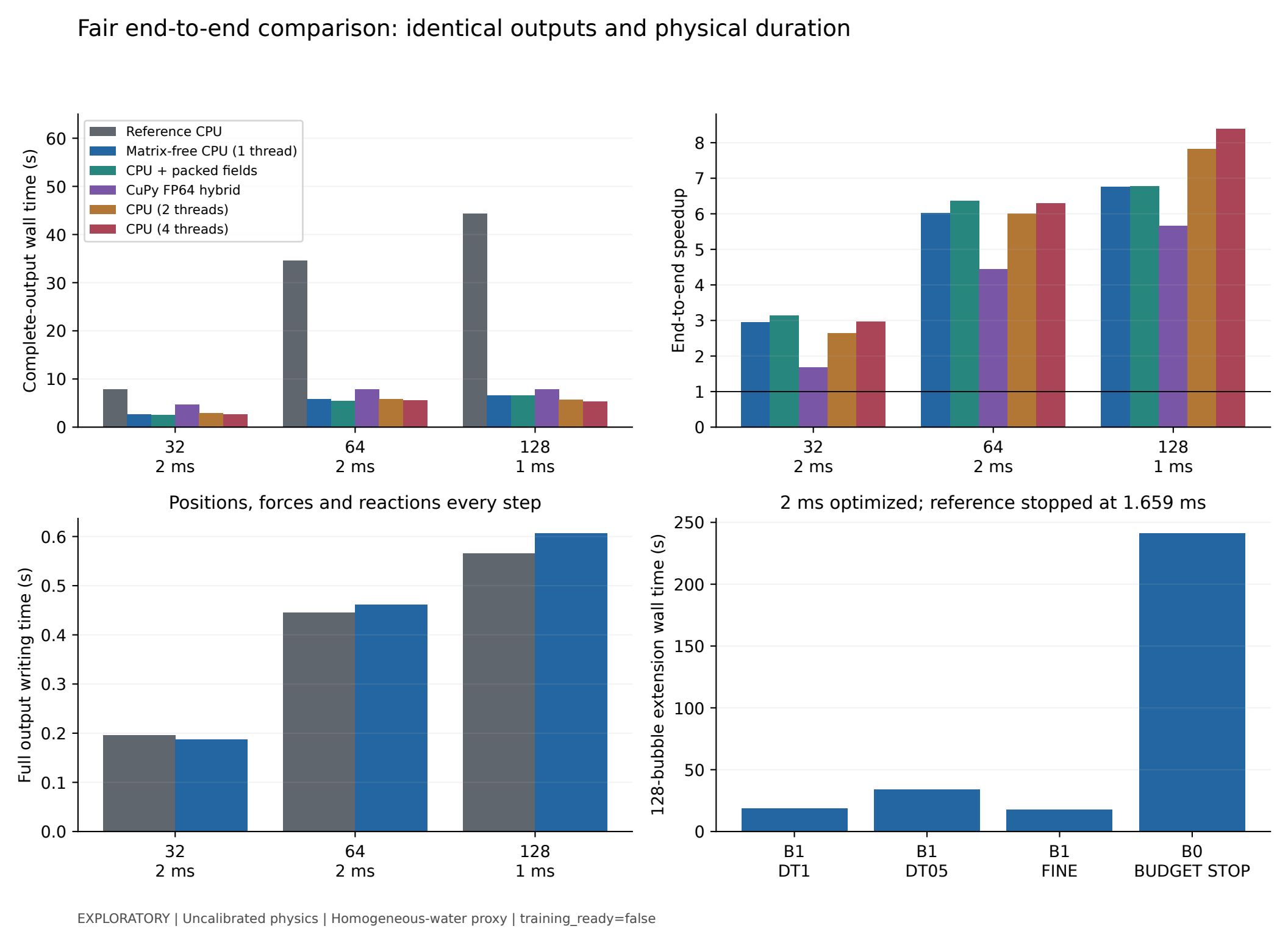
Fair end-to-end comparison: identical outputs and physical duration
Reference CPU
Matrix-free CPU (1 thread)
CPU + packed fields
CuPy FP64 hybrid
CPU (2 threads)
CPU (4 threads)
Complete-output wall time (s)
0
10
20
30
40
50
60
32
2 ms
64
2 ms
128
1 ms
End-to-end speedup
0
1
2
3
4
5
6
7
8
32
2 ms
64
2 ms
128
1 ms
Positions, forces and reactions every step
Full output writing time (s)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
32
2 ms
64
2 ms
128
1 ms
2 ms optimized; reference stopped at 1.659 ms
128-bubble extension wall time (s)
0
50
100
150
200
250
B1
DT1
B1
DT05
B1
FINE
B0
BUDGET STOP
EXPLORATORY | Uncalibrated physics | Homogeneous-water proxy | training_ready=false